

# HDOC Day 37 - C Programming

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## Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

## C Program

```
#include <stdio.h>

void toUppercase()
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf("%s", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }

    printf("Uppercase string: %s\n", str);
}

void reverseString()
{
    char str[100];
    int length = 0, i;

    printf("Enter a string: ");
    scanf("%s", str);

    while (str[length] != '\0')
        length++;

    printf("Reversed string: ");
    for (i = length - 1; i >= 0; i--)
        printf("%c", str[i]);

    printf("\n");
}

void checkPalindrome()
{
    char str[100];
    int length = 0, i, palindrome = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    while (str[length] != '\0')
        length++;

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            palindrome = 0;
        }
    }
}
```

```

        break;
    }
}

if (palindrome == 1)
    printf("The string is a palindrome.\n");
else
    printf("The string is not a palindrome.\n");
}

int main()
{
    int choice;

    do
    {
        printf("\n----- STRING OPERATIONS MENU ----- \n");
        printf("1. Convert lowercase string to uppercase\n");
        printf("2. Reverse a string\n");
        printf("3. Check if a string is a palindrome\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: toUppercase(); break;
            case 2: reverseString(); break;
            case 3: checkPalindrome(); break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

# Execution Screenshot

```
dhruvpandey@DHruVs-MacBook-Air ~ % clang code.c -ocode.c
dhruvpandey@DHruVs-MacBook-Air ~ % ./code.c
```

```
----- STRING OPERATIONS MENU -----
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 1
Enter a lowercase string: 1
Uppercase string: 1
```

```
----- STRING OPERATIONS MENU -----
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 2
Enter a string: Dhruv
Reversed string: vurhD
```

```
----- STRING OPERATIONS MENU -----
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 3
Enter a string: Dhruv
The string is not a palindrome.
```

```
----- STRING OPERATIONS MENU -----
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit
Enter your choice: 4
Exiting program...
dhruvpandey@DHruVs-MacBook-Air ~ %
```